

Resistant starch lowers fecal concentrations of ammonia and phenols in humans.

We investigated the effect of resistant starch (RS) on markers of colonic protein metabolism. Eleven subjects participated in a randomized crossover study in which they consumed either high-RS (39 $\pm$ 3 g/d, χ^2 $\pm$ SEM) or low-RS (5 $\pm$ 0.4 g/d) diets for 3 wk. All other macronutrients were kept constant. During the high-RS diet daily excretion of fecal nitrogen increased from 1.84 $\pm$ 0.15 to 2.86 $\pm$ 0.42 g/d ($P < 0.01$) and excretion of fecal phenols fell from 9.2 $\pm$ 1.4 to 5.3 $\pm$ 0.8 mg/d ($P < 0.01$). Fecal concentrations of ammonia decreased from 397 $\pm$ 33 to 278 $\pm$ 49 microgram/g ($P < 0.01$) and phenols decreased from 69 $\pm$ 8 to 39 $\pm$ 10 microgram/g ($P < 0.001$). Daily output of urinary ammonia, urea, phenols, and total nitrogen did not change significantly, but pH decreased from 6.4 $\pm$ 0.1 to 6.2 $\pm$ 0.1 ($P < 0.05$) during the high-RS period. These results suggest that RS significantly attenuates the accumulation of potentially harmful byproducts of protein fermentation in the human colon.